TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders

Intercepts: set y = 0 to find the x-intercept; set x = 0 to find the y-intercept.

Solution of a system: the point where the lines cross. It makes both equations true.

How many? Different slopes: one. Same slope, different y-intercept: none. Same line: infinite.

1. Which point is the solution of the system $y = x + 2$ and $y = -x + 6$?

Tip: Check each point in BOTH equations.

A. (4, 2)

B. (0, 2)

C. (2, 4)

2. Student tickets cost $4 and adult tickets cost $8. A club sold 40 tickets for $240. Which system models this?

Tip: One equation for money, one for the number of tickets.

F. $8s + 4a = 240$ and $s + a = 40$

G. $4s + 8a = 40$ and $s + a = 240$

H. $4s + 8a = 240$ and $s + a = 40$

3. Use the intercepts to graph $3x + 2y = 12$. Where does the line cross each axis?

> **Tip:** Set y = 0, then set x = 0.

Set y = 0: 3x = _____, x = _____

Set x = 0: 2y = _____, y = _____

Intercepts: (____, 0) and (0, ____)